

Final Project Proposal

For our project, we're going to analyze a spring pendulum system. This system is just like a simple pendulum, but instead of a string there's a spring. Using Lagrangian mechanics, the equations of motion for small values of ϕ and ϵ can be found:

$$\phi(t) = A \cos(\sqrt{\frac{g}{l}}t - \delta)$$
$$\epsilon(t) = B \cos(\sqrt{\frac{k}{m}}t) + C \sin(\sqrt{\frac{k}{m}}t)$$

where ϕ is the angle of the pendulum from the vertical, ϵ is the difference between the equilibrium length of the spring l and the length of the spring at any given time t , δ is the phase shift, g is the acceleration due to gravity, k is the spring constant, and m is the mass of the object at the end of the spring.

To analyze this system we're going to measure the height of the mass and the angle (ϕ) over time. To do this, we're going to set up a meter stick and a protractor behind the pendulum while it's swinging. Then afterwards, we can look back at the video and take measurements at whatever time interval we desire.

After taking the data, we can then use nonlinear fitting to fit the first function to the ϕ and time data with A , l , and δ as parameters. Once we have l , we can use the ϕ , height, and time data to calculate r as a function of time and then ϵ as a function of time. We can then fit the second equation to the epsilon and time data with B and C as parameters. If we correctly propagate the error all the way through each step, we should have two equations that adequately describe this system.

The materials list includes a spring, a mass, a meter stick, a protractor, and a video recording device (a cell phone should work). A rough timeline of how we expect the project to go is included below:

- Day 1: Gather materials, work on a program for analyzing the data.
- Day 2: Collect and compile height and angle data. Work on the lab report.
- Day 3: Analyze the data collected the previous day. Work on the lab report.
- Day 4: Adjust the program if needed. Work on the lab report.
- Day 5: Collect and compile height and angle data. Work on the lab report.
- Day 6: Analyze the data collected the previous day. Work on the lab report.
- Day 7: Put final touches on the lab report.